

source of that sensitivity is a 5mm², three-layer silicon chip. A portion of the centre wafer is suspended between the two outer wafers by flexible silicon beams. When the chip moves, the suspended centre lags behind due to its inertia. A measurement of that relative motion is used to calculate the speed, direction and distance the chip has moved. This exquisitely sensitive accelerometer can detect a 10-femtometre change in the position of its centre chip. As a result, it can measure changes to acceleration in the micro-gravity range. That's about 1000 times more sensitive than accelerometers used in a Wii, an iPhone or an automobile's airbag system."

Another interesting development is use of nanomaterials to boost a standard chemical and biological detection technology (Raman spectroscopy) to 100 million times its usual sensitivity rates. As sensitivity rises, sensor size can shrink. That could lead to detectors small enough to clip onto a mobile telephone. "With a wave over produce, the sensor might warn consumers of salmonella on spinach leaves or pesticides present in 'organic' produce," Hartwell says in the press report. It adds the senses of taste and smell to CeNSE.

The team is simultaneously working on adding various other sensors ranging from light and pressure to temperature and humidity.

iPhone turned sensor

It seems to be the season for mobile-sensor convergence, and it is not just hobbyists experimenting with such clip-onto-mobile sensors. Jing Li, a physical scientist at NASA's Ames Research Center, has developed a technology to add low-cost, low-power, high-speed nanosensor-based chemical sensing capabilities to cellphones.

The stamp-sized device can be plugged into an iPhone's 30-pin dock connector to collect, process and transmit sensor data. It is capable of detecting and identifying low concentrations of airborne ammonia, chlorine gas

and methane. According to the NASA report, the device senses chemicals in the air using a 'sample jet' and a multi-channel silicon-based sensing chip which consists of 16 nanosensors, and sends measured data to another phone or a computer via a telephone or Wi-Fi network.

Sensor network for aircraft monitoring

A team at Scotland's Institute for System Level Integration (iSLI) has started designing a wireless sensor system that could well become a standard feature of the next generation of commercial aircrafts. The project has been launched with a budget of 3.3 million pounds, with contributions from various aerospace industry leaders.

The first prototype wireless sensor network, WITNESS, will gather complex and accurate data from different parts of the aircraft, which will be used to help identify technical faults, optimise performance and monitor the overall health of the aircraft.

Wireless is a key capability to reduce the costs associated with wired sensor cables. According to Dr Begbie, director, iSLI, "Putting in cables adds to aircraft cost and, crucially, weight and it's not the easiest thing to get in and maintain. If you take Rolls-Royce as an example, when it is developing a new engine it can have upwards of 3000 sensors attached when it is on the testbed. Routing all the wires to the sensors and bringing them all back to a collection point is a big job, and when you have 3000 cables running over a vibrating engine, you get a lot of difficulty with drop-outs. So Rolls-Royce wants us to look at how the wireless technology can help."

Subsequently, the system can be improved to take on tasks like predictive maintenance and, ultimately, real-time data for safety-critical components. It may also be of great use in the carbon fibre components being developed to replace aluminium aircraft parts, where sensors can be used to help monitor how the components are

copied with high pressures and heavy loads and allow manufacturers to get better information about super-structural capacity and the life expectancy of each component.

Wearables to fight flab

According to a recent *New York Times* report, scientists are developing wearable wireless sensors to monitor overweight and obese people as they go about their daily lives.

Made of sophisticated sensing instruments, the devices will track how many minutes the wearers work out, how much food they consume, whether they went for the routine walk at the park or sneaked into a fast-food joint, and so on. The goal is to make the users aware of where, when and how they added more calories to the existing pile—or, on a more positive note, how they managed to lose some of it!

Coming up next...

As far as the field of wireless sensors goes, we have seen only the tip of the iceberg.

The next generation might live a life completely controlled by sensors—sensors that detect how much they eat, how many calories they burn, by how many micrometres their waistline expands or shrinks in a day, how well their faculties and organs are functioning, how many minutes they work, how well their car is working, when and how many lights are turned on in their home, what volume of pollutants they add to the air, how much electricity they consume in an hour, how long they use each device and what not.

Some day, the bubble will burst as security and privacy issues loom large, but just as the World Wide Web emerged stronger and better after the dotcom bust, sensors too will emerge victorious and perhaps really make the whole Earth an Internet of Things. ●

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